

Machine Learning Algorithms and Models

The machine learning landscape includes numerous algorithms and models, each with distinct characteristics and applications. Linear regression fits linear relationships between inputs and continuous outputs. Logistic regression performs binary classification using sigmoid functions. Decision trees recursively partition data based on feature thresholds, creating interpretable models. Random forests combine multiple decision trees, improving robustness and reducing overfitting. Support Vector Machines (SVMs) find optimal hyperplanes separating classes in high-dimensional spaces. K-Nearest Neighbors classifies points based on neighbors, making lazy learning decisions at prediction time. Neural networks stack layers of interconnected nodes, enabling complex non-linear function approximation. Convolutional Neural Networks (CNNs) excel at image processing through weight sharing and local connectivity. Recurrent Neural Networks (RNNs) and transformers process sequential data like text or time series. Gradient Boosting builds ensemble models sequentially, with each new model correcting previous errors. Naive Bayes applies probabilistic reasoning assuming feature independence. K-Means clustering partitions data into k clusters minimizing within-cluster variance. Gaussian Mixture Models assume data comes from multiple Gaussian distributions. Generative Adversarial Networks (GANs) pit generator and discriminator networks against each other, enabling realistic synthetic data generation. Selection depends on problem type, data characteristics, and computational constraints.